

CREATIVITY, INC.

Overcoming the Unseen Forces that Stand in the Way of True Inspiration

BY ED CATMULL (with Amy Wallace)



The Big “So What”

Being creative and building a creative culture is a complex subject. Ed Catmull, President of Pixar Animation and Disney Animation, brings the process to life by sharing what happens behind-the-scenes in these 2 great companies.

Catmull believes **there's no fixed formula for creativity, and innovation management is an ongoing, never-ending job.** By sharing his reflections, mistakes, and the approaches used to remove barriers to creativity, Catmull shows us **how creativity really works, and how we can likewise build and maintain a creative, solutions-oriented environment in our organizations.**



Introduction

The book is written in a chronological order in 4 parts - Getting Started, Protecting the New, Building and Sustaining, and Testing What We Know - with key learning points incorporated in each section. **These tips are not success formulas, but merely starting points for you to think about, and experiment with, to find what works for you.**

In this summary, we'll present the key ideas in 3 parts:

- Key milestones;
- Foundations and barriers to creativity; and
- Building a creative culture.



Part 1: Key Pixar Milestones

In this section, **we'll zoom in on some of the key developments that shaped Pixar and Disney Animation.**



A dream was born. Since childhood, Catmull has aspired to be a Disney animator, but took the path of a computer scientist

KEY QUOTES

“This book... is for anyone who wants to work in an environment that fosters creativity and problem solving.”

“My goal has never been to tell people how Pixar and Disney figured it all out but rather to show how we continue to figure it out, every hour of every day. How we persist.”

“My job as a manager is to create a fertile environment, keep it healthy, and watch for the things that undermine it.”

“Figuring how to build a sustainable creative culture... wasn't a singular assignment. It was a day-in-day-out, full-time job. And one that I wanted to do.”

instead. At the University of Utah, he experienced what it was like to be given all the resources and freedom to pursue his passions. By the end of his PH.D program in 1974, he had several innovations under his belt, and had crystallized his goal to make a computer-animated film - something unheard of at that time, when animation was 100% hand-drawn.



Catmull was hired to head the new computer graphics at the New York Institute of Technology, and was given full autonomy to assemble his team. Despite his fears, he hired Alvy Ray Smith, whom he felt was more qualified than him. Smith became one of Catmull's closest friends and collaborators, and he learned this key lesson: **Always try to hire those who're smarter than you.**



Pixar is Born. After the Star Wars movie was launched in 1977, George Lucas hired Catmull to head the Lucasfilm computer graphics division, to integrate video editing and digital audio into movies. After 4 years, the "Pixar Image Computer" (PIC) was born – it could scan live-action footage, add special effects images, and record the combined output back onto film.



Storytelling. In 1983, Catmull met animator John Lasseter, who later lost his job at Disney and joined Pixar. With a storyteller onboard, they created their first production. Although the team ran out of time and had to launch with some incomplete wire-frames scenes, , the audience were so fascinated by the story they didn't notice the flaws. They learned another key lesson: **technical perfection is less important than the right story that touches hearts.**



Pixar, a Hardware Company. In the same year, Lucas decided to streamline the business, including selling Pixar. In 1986, Steve Jobs bought majority shares in Pixar, and Catmull suddenly found himself as the President of a hardware company selling imaging computers - a role he had no experience in. To learn, Catmull read widely and consulted experts for advice.

- He soon found that the ideas in most books sound good, but hold little practical value, e.g. "have a singular focus" sounds logical, but doesn't tell you what to focus on. "Expert advice" to set high prices for the PICs only created an expensive image that they couldn't reverse. **In the search for simple answers on how to price their product, they'd missed the more vital questions**, like how to meet customers' expectations and how to develop software that make PIC.

KEY QUOTES

"By ignoring my fear, I learned that the fear was groundless."

"Always take a chance on better, even if it seems threatening."

"For all the care you put into artistry, visual polish frequently doesn't matter if you are getting the story right."

"A conversation with Steve took you places you didn't expect. It forced you not just to defend but also to engage."

"There is nothing quite like ignorance combined with a driving need to succeed to force rapid learning."

"Not everything is simple, and to try to force it to be is to misrepresent reality."

- Catmull was also heavily influenced by Edward Deming's teachings - to empower employees at all levels to find and fix problems. This would become a key foundation at Pixar.



For years, Pixar was bleeding cash. Finally, in 1991, it secured a **Disney contract**, where Disney would fund, own and distribute 3 Pixar movies. Lasseter assembled a great team for Toy Story, but the production process was full of challenges and conflict with Disney. Nearing the launch, Jobs predicted that with Toy Story's success, Disney would see Pixar as a potential rival. To safeguard Pixar's position, he decided to **take Pixar public**. In 1995, Toy Story broke box office records, the IPO raised almost \$140m, and Jobs renegotiated a 50-50 split with Disney for future movies.



Forging Pixar's identity. In 1997, Pixar decided to let a new team drive Toy Story 2, while the original team worked on A Bug's Life. More than a year later, they realized Toy Story 2 was too far off the mark. Although Disney gave the green light, they decided to redo the film. The gruelling 9 months that followed shaped Pixar's identity in many ways:

- Despite their guiding principles - "Story is King" and "Trust the Process" - Pixar had created the wrong story and missed the gaps. Principles weren't enough; they had to stay alert to changes.
- By refusing to accept lower-quality releases, they affirmed their commitment to quality.
- To fix the film, people came together to constructively critique the story - this evolved organically into Braintrust, a powerful feedback mechanism elaborated in Part 3.
- They learned that great ideas, in the hands of the wrong people, would flop. However, if you give the right team (right talents with chemistry) the support and resources they need, they'll fix or recreate any idea to make it great.
- To launch Toy Story 2, everyone worked 7 days a week and barely saw their families. An exhausted artist even left his infant in the car and returned to find him unconscious. By the launch, people were totally burnt out. Pixar learned that, for long-term productivity and happiness, you need a balanced approach. For Pixar, this means making it easy for

KEY QUOTES

"You don't have to ask permission to take responsibility."

"Being on the lookout for problems... was not the same as seeing problems."

"Everyone says quality is important, but they must do more than say it. They must live, think and breathe it."

"If you give a good idea to a mediocre team, they will screw it up. If you give a mediocre idea to a brilliant team, they will either fix it or throw it away and come up with something better."

"If you get the team right, chances are that they'll get the ideas right."

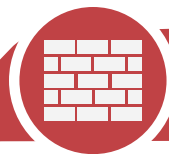
staff to enjoy courses and facilities like swimming pools at the office, and not penalizing those with families for putting in less time at work.



A new goal. With Toy Story's success in 1995, Catmull finally fulfilled his goal of making a computer-animated film. However, as new problems emerged at Pixar (including those above), Catmull found his new calling – to learn what it takes to build a lasting creative culture, and build Pixar to outlive the forces that've brought down other great companies. We'll cover some of these philosophies and mechanisms in Parts 2 and 3 of this summary.



In 2006, Pixar was sold to Disney for \$7.4 billion, and Catmull and Lasseter became President and Chief Creative Officer respectively for both Disney Animation and Pixar. They could now test if Pixar's approach could be replicated at Disney Animation, while preserving Disney's unique heritage. After their fair share of victories and mistakes, a revitalized Disney Animation launched Tangled in 2010 - their first #1 hit in 16 years.



Part 2: Foundations & Barriers to Creativity

In this segment, we'll highlight the key ingredients for creativity and problem-solving, and the barriers that stand in their way. In Part 3, we'll address how to overcome these barriers.

Discovery & Uncertainty



Many people think that a creative breakthrough starts with a brilliant idea by a visionary, who leads his people to victory. **The truth is, most "geniuses" discover their vision through struggles over time, figuring things out along the way.**

- In Part 1, we had a glimpse of the long journey from Catmull's initial goal to the completion of Toy Story 20 years later.
- Monsters, Inc. started out as a story of a 30 year old man coping with scary characters (that only he could see). The final product 5 years later was a markedly different movie about monsters who scared kids for a living.
- Director Pete Docter knew what he wanted to express in the movie, Up, but he took years to conceptualize a story that could

KEY QUOTES

"It was management's job to take the long view, to intervene and protect our people from their willingness to pursue excellence at all costs."

"The act of thinking about the problem and responding to it was invigorating and rewarding."

"At the heart of this merger was an understanding that both companies had to trust each other."

"Creative people discover and realize their visions over time and through dedicated, protracted struggle. In that way, creativity is more like a marathon than a sprint."

"We humans like to know where we are headed, but creativity demands that we travel paths that lead to who-knows-where."

"The best way to predict the future is to invent it."
- Steve Jobs

bring his vision to life. It started out as a floating castle with a king and his 2 sons vying for the throne. Four major iterations later, only the title and a tall bird from the original story remained.

In short, discovery means you must embark on the process without knowing the answer. This is uncomfortable for most people.

Getting Uncomfortable: Candour, Failure, Change



Candour. To create breakthroughs, candid feedback is invaluable. However, there are many reasons for people not to be candid at the workplace, e.g. fear of saying something stupid, offending others, or jeopardizing their career.



Failures and mistakes are also inevitable as part of trying something new. Yet, most of us have been conditioned to see failure as something bad and shameful, and we'd avoid it if possible. Organizations make it even worse by penalizing people for mistakes.



Change. It's natural to want to hang on to what works. Yet, as things change, we can't really know for sure what still works and what should be discarded. Most people also subconsciously resist change as they are worried (a) about the work involved, (b) to be seen as weak-minded or not knowing what they're doing; and/or (c) to preserve self-interest. Change is unavoidable. Denying the need for change won't stop its eventuality.

Blind Spots and Mental Models



The Unknown. Imagine you're standing on 1 side of a door, which represents the world you see and know. The other side is the endless world of the unknown and unseen. Staying on this side of the door feels secure, but in reality, the unknown forces on the other side are at work whether you see them or not.

- Most people don't know that Catmull had a close shave with death when he was 12 - had things gone differently, Pixar (including the people it touched and the events it triggered) won't exist. Likewise, our circumstances are shaped by countless hidden factors and events that we're oblivious to.
- Hierarchy (which is necessary in organizations) can also limit access to crucial information. As you move up the ranks,

KEY QUOTES

"The process of coming to clarity takes patience and candor."

"By definition, 'discovery' means you don't know the answer when you start."

"Without the critical ingredient that is candor, there can be no trust. And without trust, creative collaboration is not possible."

"Failure isn't a necessary evil. In fact, it isn't evil at all. It is a necessary consequence of doing something new."

"Fear makes people reach for certainty and stability, neither of which guarantee the safety they imply."

"People want to hang on to things that work... as we become successful, our approaches are reinforced, and we become even more resistant to change."

"To be truly humble... leaders must first understand how many of the factors that shape their lives and businesses are - and will always be - out of sight."

people tend to “manage upward” to present only what you’d like to see and hear, distorting the true picture of what’s happening. People in *technical or front-line roles* also don’t generally surface issues - they assume you already know, there’s no need to involve upper management, or they try to appear in control.

 **Randomness.** Our brains are designed to look for patterns, even when they don’t exist. When a company does well, we tend to explain it using “success factors” or “organizational strengths”, which are often over-simplifications of the truth. The media further reinforces these myths.

 **Mental Models.** Besides seeing incomplete pictures and false patterns, we also assume (wrongly) that we see clearly with *hindsight*. The truth is, our brains fill in gaps with assumptions and patterns (without our awareness), and we’re biased toward memories that hold meaning for us. That’s why different people see and recall an event differently. Yet, **we typically assume that our view is the best and correct one. Our mental models can distort our worldview, and make us rigid or blind to alternative solutions.** This problem is compounded at an *organizational level*, as people’s mental models become intertwined when they work closely together. Organizations develop biased mental models based on their know-how and experiences.

Leaders must remember that there are many hidden and random factors affecting their success, to be humble to alternative views, and constantly seek to uncover their blind spots.

Protect the New



A large organization needs projects and revenues to keep its staff and facilities occupied. More revenues lead to more staff and pressures to grow, creating a self-perpetuating cycle (“**feeding the beast**”). Ultimately, standards drop. On the other hand, new creations are like **ugly babies** – they look awkward, and you don’t know if they’ll grow up, much less how they’ll look.

For creative solutions to blossom, we must protect the Ugly Baby (new ideas) from the Hungry Beast (deadlines and targets). This is not to preserve old practices or interests, but to prevent new concepts from being killed prematurely. For instance, Pixar wanted to introduce internships, but the departments didn’t

KEY QUOTES

“We ignore the things we cannot see and make deductions and predictions accordingly.”

“Because (leaders) weren’t aware of these blind spots, they assumed that the problems didn’t exist.”

“We must be careful not to believe our own hype.”

“We are aware of the results of our brain’s processing but not the processing itself.”

“Most of us walk around thinking that our view is best – probably because it is the only one we really know.”

“If you don’t try to uncover what is unseen and understand its nature you will be ill prepared to lead.”

“When someone hatches an original idea, it may be ungainly and poorly defined, but it is also the opposite to established and entrenched – and that is precisely what is most exciting about it.”

“Our job is to protect our babies from being judged too quickly. Our job is to protect the new.”

want to pay for inexperienced interns. Pixar hired the interns using corporate budget, and made them available as a shared resource. The program turned out to be so successful that the departments began to pay for their own interns.



Part 3: Building a Creative Culture

In this segment, we'll highlight some of the mechanisms used at Pixar to overcome the barriers to creativity. Combined, these mechanisms build candour, make it okay to make mistakes, challenge mental models and uncover truths, and tap on the brainpower of every team member.

Broadening our Viewpoints



Rather than pretend or try to know everything, it's better to accept that we *can't*. Instead, **focus your attention on how to combine differing viewpoints, treating them as additive rather than competing.** Here are several mechanisms that Pixar uses to expand perspectives:



"Dailies" are problem-solving sessions where team members share their work-in-progress, get constructive feedback, and get applauded for their work. This makes it safe to show incomplete work, give/receive feedback, discuss problems, learn from and inspire others.



Research Trips. Pixar teams immerse themselves in the subjects of the film, e.g. the Ratatouille team spent 2 weeks in France dining in Michelin-starred restaurants, learning how chefs work and even trudging through sewers where rats live. Such trips challenge clichés, bring fresh inspiration, and allow the teams to bring realistic microdetails that make the movies rich and authentic.



Useful limits. Creative work can be like bottomless pits. Setting limits force us to (a) find smarter ways to do things, (b) stop trying to perfect things, and (c) prioritise finite resources. When setting limits, focus on empowering people to solve problems, rather than preventing them from screwing up.



Combine different fields. The integration of art and technology is vital at Pixar. It challenges people to step outside their specialization, review their mental models, and has inspired many innovations, like the "Review Sketch Tool" (which allows you to sketch suggestions onto a digital image), and the "Pitch Doctor" (to do a realistic pitch for feedback).

KEY QUOTES

"The cost of preventing errors is often far greater than the cost of fixing them."

"Inspiration can, and does come from anywhere."

"If there is fear, there is a reason – our job is to find the reason and to remedy it."

"Research trips challenge our preconceived notions and keep clichés at bay. They fuel inspiration."

"Any time we impose limits or procedures, we should ask how they will aid in enabling people to respond creatively."

"Our specialized skills and mental models are challenged when we integrate with people who are different."



Small experiments. In 1996, Pixar's revived its short films (screened for free before movies) as part of its R&D efforts. Although the "shorts" can cost millions to produce, they allow Pixar to test new technologies, approaches, or talents before deploying them for bigger projects (thus reducing failure risks). This also creates opportunities for smaller teams to bond, connect with moviegoers, and sends an internal signal that Pixar values artistry.



Learn to see. Pixar's drawing program teaches more than drawing skills. To draw objects as they are (not what you *think* they are), the staff must become aware of their preconceived notions and learn to put them on hold. Drawing variations (e.g. upside-down objects) also hone observation skills.



Post-mortems help (a) create opportunities for reflection, (b) consolidate what's been learned, (c) teach others who weren't present, (d) address misunderstandings, and (e) equip future project teams with the right questions to ask. However, people generally dread post-mortems. To make them effective:

- o Vary the format, and use mid-mortems to identify topics to focus on;
- o Get people to list the top 5 things they'd want to do again and 5 things that they wouldn't. This softens the approach.
- o Use data as part of your analysis.



Continuous learning. Pixar offers a wide range of classes e.g. painting, acting, belly-dancing. This creates opportunities for employees to meet and bond with new people. It also forces people to try new things, and get comfortable with imperfections.

Braintrust Meetings



In creative work, the creators must immerse themselves in the projects, yet not miss the forest for the trees. Braintrust meetings have been a valuable source of constructive feedback for Pixar directors. The meeting starts with a brief intro by the director, who then shows his work-in-progress and invites feedback. **Braintrust is unique in 2 key ways:**



It's made up of qualified participants who're well-versed in the subject matter (which, in Pixar's case, is storytelling). It's like

KEY QUOTES

"The goal is to learn to suspend, if only temporarily, the habits and impulses that obscure your vision."

"A good postmortem arms people with the right questions to ask going forward."

"Think of data as one way of seeing, one of many tools we can use to look for what's hidden."

"The attempt to avoid failure... makes failure more likely."

"Seek out people who are willing to level with you, and when you find them, hold them close."

having a panel of specialists to advise you on a medical case. People may argue passionately, but the focus is *always* on the project. There's no hidden agenda, and nothing is taken personally.



There's no hierarchy/ authority - everyone is treated as equals, with the aim of surfacing potential issues to help their colleague succeed. The creator can choose to adopt any or none of the feedback.

Disney Animation successfully adapted its Storytrust using this approach, and Catmull believes any organization can create its own Braintrust.

Creative Experimentation



In science experiments, every outcome is good, as it provides data to support or disprove a hypothesis, and unexpected discoveries may emerge. It's the same with creative experimentation.



Identify your Red Flags. A key challenge is to balance creative and business needs, and to know where to draw the line. For Pixar, the red flag is when a director loses the crew's confidence. As a director-led company, when team members feel lost and confused, it's a sign the leader has lost direction, and the project is in trouble.



When failures occur, it's important to learn from them. At Pixar, if major issues surface, it's common for the key personnel to dedicate 2 days for an offsite meeting, to focus on understanding the problem and its root cause(s). At such meetings, everyone participates, and no one shirks their roles or responsibility.



It's crucial to create an environment where fear and failure are separated. When leaders respond constructively to failure, people are not fearful to make mistakes, and can work in unity to solve key problems.

Handling Change and Randomness



When directing "Up", Pete Docter used these tips to keep himself and his team motivated despite the changes and uncertainty:

- o List the problems and group them into 2-3 broader issues -

KEY QUOTES

"Frank talk, spirited debate, laughter, and love. If I could distill a Braintrust meeting down to its most essential ingredients, those four things would surely be among them."

"The key is to look at the viewpoints being offered... as additive, not competitive."

"To be a truly creative company, you must start with things that might fail."

"If the crew is confused, then their leader is, too."

"Management's job is not to prevent risk but to build the ability to recover."

"It is not the manager's job to prevent risks. It is the manager's job to make it safe to take them."

"Working with change is what creativity is about."

it's easier to tackle a handful of issues than feel everything is wrong.

- o Instead of announcing a change, “trick” people to chat about it. When they catch the idea or vision, they’ll be excited to make the change.
- o If you’re stuck, take a break to play and have fun. Return to tackle the issue when your mood has shifted.



When random mistakes happen, there’s no point in blaming the staff. Focus on the solution and future safeguards.



We tend to draw an artificial line between “small” vs “critical” problems, and treat them differently. It’s better to treat all problems equally, and empower people to solve problems – this prevents small problems from compounding, and people are less likely to freeze in fear when they face a big issue.



Recognize that human potential is random and can’t be predicted. We can only find out over time who will step up to the challenge.

Constructive Mental Models



With so many obstacles to overcome, it’s normal to feel overwhelmed. At Pixar, people use familiar pictures (or mental models) to handle fear, challenges and tough decisions. Here are some examples:



Directors and creative roles often involve pursuing an unseen destination. Pete Docter compares directing to running through a long tunnel – you may not see your way, but you know an opening exists. Brad Bird uses skiing to remind himself not to over-analyze; the only way to go fast without crashing is to relax. George Lucas sees his company like a train full of team members who’re determined to get to their destination. Bob Peterson thinks of making a movie like an archaeological dig – you uncover pieces each time, and finally get an idea of what it is.



Producers and logistical roles require a different set of models. John Walker imagines balancing an upside-down pyramid perfectly on the palm of his hand. Lindsey Collins imagines herself as a chameleon who can take on the personality needed for the task at hand. Katherine Sarafian imagines people on a

KEY QUOTES

“While we can agree in principle that an organization needs to be flexible in order to solve problems, living up to that principle can be extraordinarily difficult.”

“Big and small problems are, in key ways, the same.”

“Everyone says they want to hire excellent people, but in truth we don’t really know, at first, who will rise up to make a difference.”

“I’ve known many people I consider to be creative geniuses... yet I can’t remember a single one who could articulate exactly what this vision was that they were striving for when they started.”

“Our mental models aren’t reality. They are tools... The tool is not reality. The key is knowing the difference.”

different floors of a condo (each with a different view, and can take the lift to visit others).



Catmull personally finds mindfulness practice to be useful in management, as it helps him to focus on the problem calmly without panic nor avoidance.

Notes Day



While Catmull and Lasseter were transforming Disney Animation, Pixar was starting to face its own crisis. After years of success, people were starting to *self-censor*. Meanwhile *production costs* were increasing, with growing *external market pressures*. At an offsite retreat, Pixar leaders identified the need to reduce costs by 10%. They decided to close down the office for 1 day, so everyone could jointly solve the problem. Notes Day was born.



Leaders volunteered to run the event, the goals were explained to staff over 3 sessions, and electronic suggestion boxes were set up. Some 4000 emails (with ideas on topics and how to run Notes Day) were screened and distilled to 120 topics (e.g. Tools & Technology, Environment & Culture). Sessions were created based on the topics people wanted to sign up for, and facilitators were trained to run the sessions to extract proposals, ideas, best practices, and next steps.



On that day, Lasseter set the tone by sharing candidly how hard but vital it was for him to receive 2.5 pages worth of personal feedback, and encouraged everyone to embrace the process. By the end of the day, Pixar was abuzz with excitement. People had met new colleagues, understood problems and successes in other departments, and felt deeply invested in Pixar's future.



Notes Day's success was due to at least 3 factors: There was a clear goal (cut costs by 10%), it was championed by the highest levels of the company, and was led and organized from within. More importantly, it demonstrates that sustaining a creative culture is an ongoing effort that calls for constant adjustments and innovations.



Other Details in the Book to Look out For

Stories are powerful, and this book is packed with stories that help you to experience the challenges, triumphs, and learning experiences

KEY QUOTES

"To keep a creative culture vibrant, we must not be afraid of constant uncertainty."

"The future is not a destination – it is a direction. It is our job, then, to work each day to chart the right course and make corrections when, inevitably, we stray."

through Catmull's eyes. We've captured some of the key highlights and tips in this summary, though there's much more to be gleaned from the details in this inspiring and engaging book, including:

- Behind-the-scenes processes, struggles and fixes for movies like Toy Story, Finding Nemo, Tangled, Ratatouille, Bolt etc;
- Deliberations and decisions that give us a glimpse of great minds (like Jobs, Lasseter, Lucas, Catmull etc.) at work;
- The challenges and steps taken to manage Pixar and Disney Animation after the 2006 merger; and
- An afterword about Steve Jobs and a list of "starting points" you can use to build a creative culture.

KEY QUOTES